

GALVI 315 interface study - design brief

AI BRAKING / AB-G315 / C0 / 2026-09-07

Original integration geometry referencing GALVI N.315.HYD.030/05 and PD.315 catalogue dimensions. This is not a GALVI manufacturing model or verified replacement. Purchased brake internals, load paths and shaft locking are not released.

TARGETS (not validated)

Drum outside diameter: 315 mm. GALVI PD.315; PDF page 15

Brake axis height: 230 mm. GALVI N.315; PDF page 4

Four mounting holes: Ø18; 440 × 80 mm pitch. GALVI N.315; PDF page 4

Drum band width: 118 mm. GALVI PD.315 B1; PDF page 15

Finished bore selected: 50 mm nominal. Our selection within published R max 80; shaft requirement pending

DESIGN DECISIONS

Buy or partner for the complete brake and actuator; focus our first integration effort on the mounting and drum/shaft interface.

Use published dimensions for external interfaces only. Keep proposed section thickness and shaft selection visibly separate.

Request a current GALVI drawing before establishing compatibility or manufacturing release.

CANDIDATE MATERIALS

GALVI publishes EN-GJL-250 and EN-GJS-500-7 drum variants; neither is selected for production here.

Our adapter plate: S355J2+N candidate, 20 mm illustrative thickness; no strength or bolt-joint approval.

Assembly and engineering gates

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ASSEMBLY INTENT

- 1. Confirm source revision and purchased brake option with supplier.
- 2. Inspect 440 x 80 four-hole pattern and datum plane on proposed adapter.
- 3. Place the brake so its drum axis is 230 mm above its mounting plane.
- 4. Confirm PD.315 axial position and shaft attachment from machine and supplier drawings.
- 5. Approve detailed joints, load paths and guarded test plan before prototype manufacture.

OPEN DESIGN RISKS

- 1. Catalogue revision February 2016; current availability and dimensions unconfirmed.
- 2. Drum wall/web geometry is an original simplification, not published manufacturing geometry.
- 3. Brake envelope contains no levers, shoes, springs or actuation mechanism; do not manufacture it.
- 4. No verified shaft connection, preload, tolerances, thermal capacity or operating rating.

VALIDATION REQUIRED

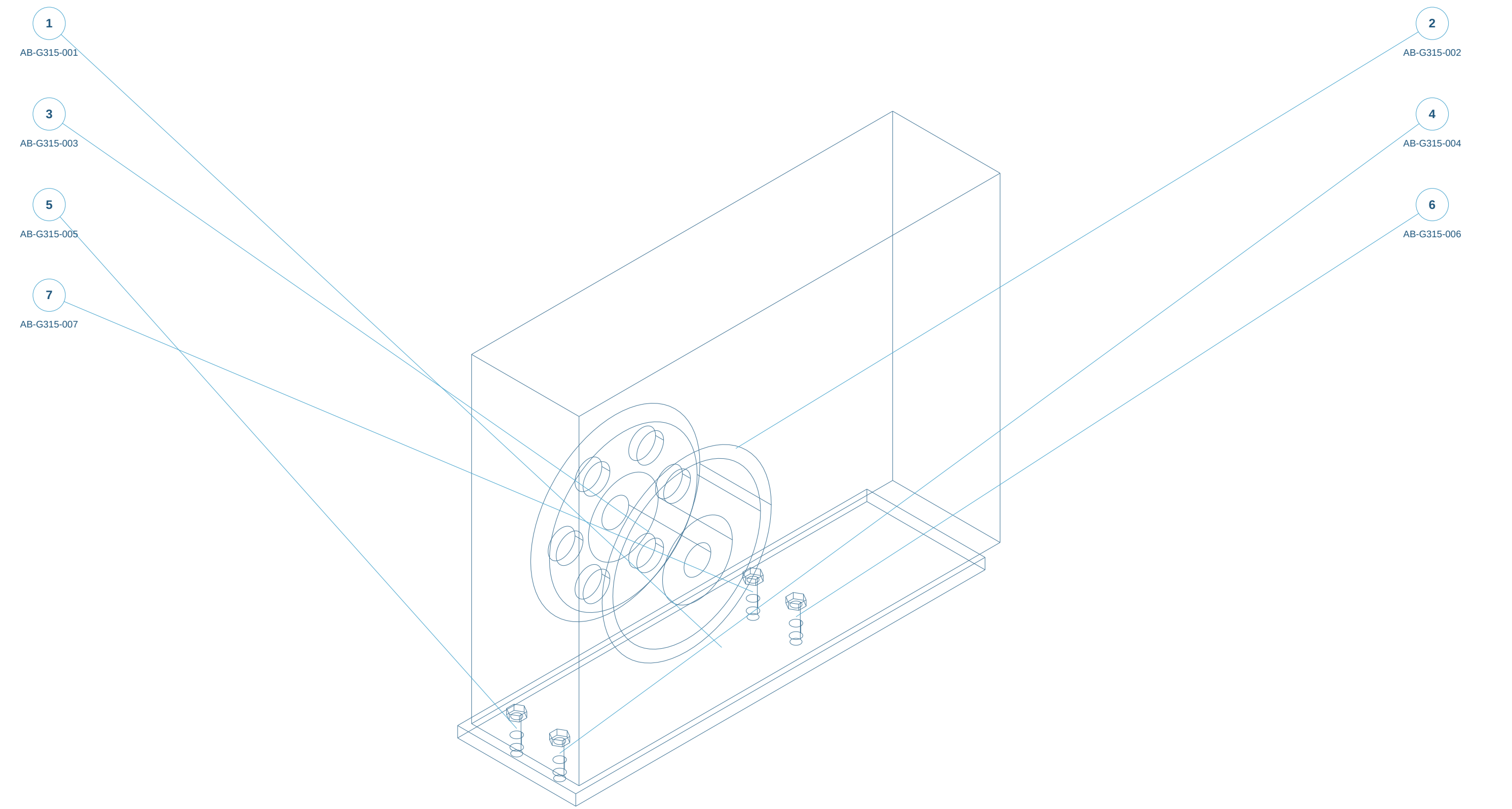
- 1. Check supplier CAD and physical first article against the published interface.
- 2. Size adapter, foundation, shaft connection and drum for approved duty.
- 3. Perform independent dimensional, torque, thermal and endurance validation.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

- 1. Request thruster supply/duty, position-switch options, wear sensing, connector and machine-safe-state requirements.

Assembly reference / numbered components

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Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

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AB-G315-001 | Our mounting adapter concept | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-G315-002 | Purchased brake installation envelope | interface-envelope

Supplier-controlled; illustrative reserved envelope; Buy complete brake; replace this envelope with current vendor STEP

AB-G315-003 | PD.315-referenced drum concept | reference-concept

GALVI material options: EN-GJL-250 / EN-GJS-500-7; selection pending; Supplier-controlled drum preferred; our simplified CAD for interface review only

AB-G315-004 | M16 mounting hardware envelope | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy after joint/preload calculation

AB-G315-005 | M16 mounting hardware envelope | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy after joint/preload calculation

AB-G315-006 | M16 mounting hardware envelope | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy after joint/preload calculation

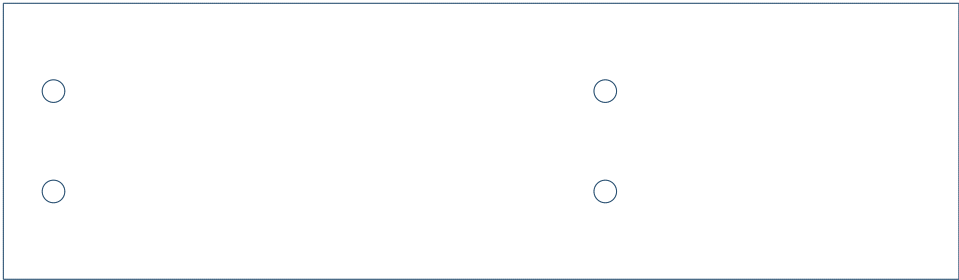
AB-G315-007 | M16 mounting hardware envelope | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy after joint/preload calculation

AB-G315-001 / Our mounting adapter concept

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 762.00 / 220.00 / 20.00 mm

4 x Ø18 THRU. Assembly X +/-220, Y +/-40; part-local X -341/+99, Y +/-40. Full pitch 440 x 80.

Plate 762 x 220 x 20 is our proposal; foundation holes, strength and flatness unapproved.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



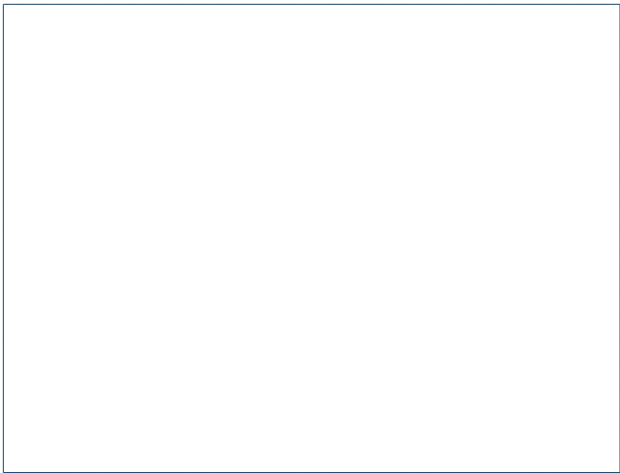
AB-G315-002 / Purchased brake installation envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy complete brake; replace this envelope with current vendor STEP

Overall X/Y/Z bounds: 784.00 / 200.00 / 595.00 mm

Published maximum extents O/P: X -244 / +540; height C=595.

200 mm transverse envelope is our assumed display width; actual full width requires vendor CAD.

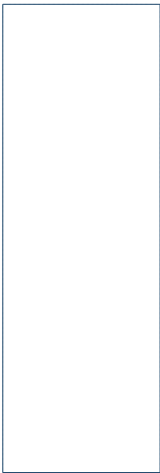
Volume is reserved space, not a physical solid to manufacture.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

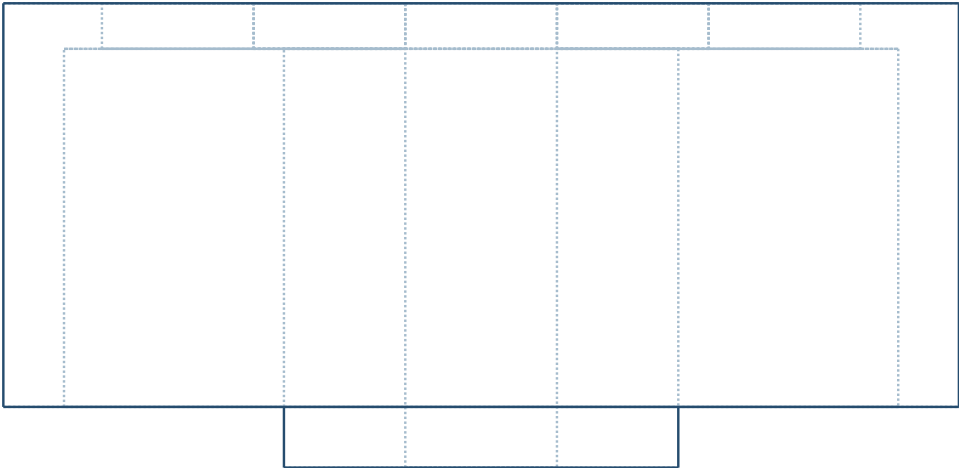
SIDE Y-Z



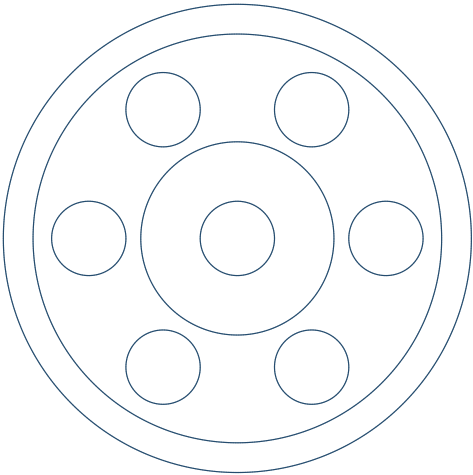
AB-G315-003 / PD.315-referenced drum concept

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TOP X-Y



FRONT X-Z



GALVI material options: EN-GJL-250 / EN-GJS-500-7; selection pending

Supplier-controlled drum preferred; our simplified CAD for interface review only

Overall X/Y/Z bounds: 315.00 / 153.00 / 315.00 mm

Published: OD 315, band 118, overall length 153, hub OD 130, 6 x Ø50 on PCD200.

Our choices: shaft bore 50, rim inner diameter 275, 15 mm flat web. Internal section is NOT GALVI geometry.

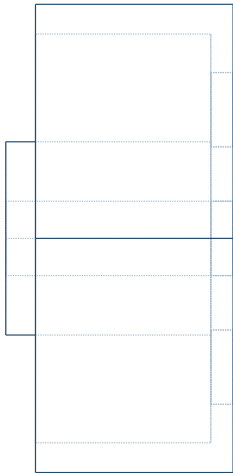
Coupling holes are pin interfaces, not six mounting-bolt holes. No speed or torque rating transferred.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

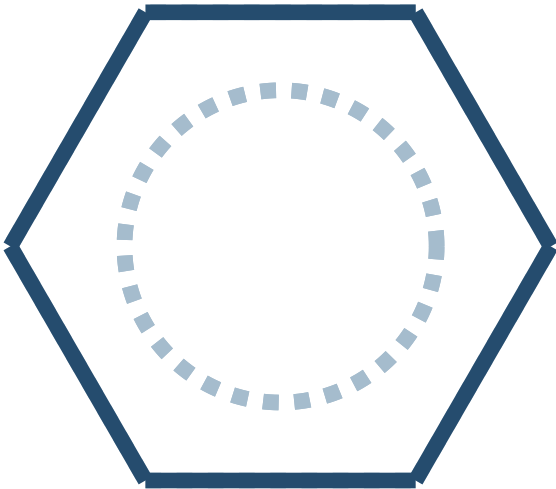
SIDE Y-Z



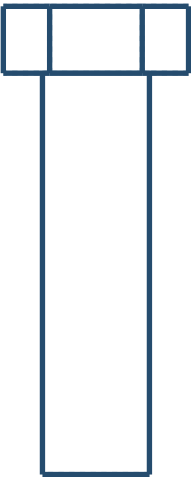
AB-G315-004 / M16 mounting hardware envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy after joint/preload calculation

Overall X/Y/Z bounds: 27.70 / 23.99 / 70.00 mm

M16 is our candidate for Ø18 clearance holes, not a GALVI specified bolt.

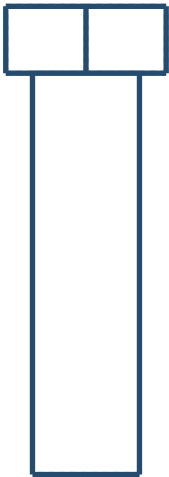
Nut, washer stack, grip, grade and preload unresolved; envelope only.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

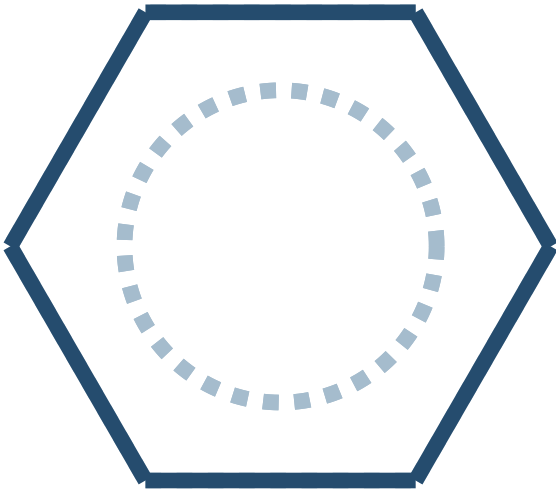
SIDE Y-Z



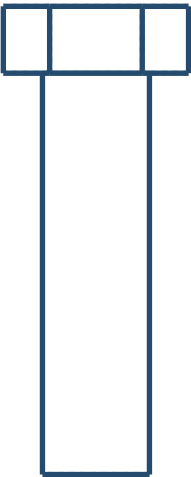
AB-G315-005 / M16 mounting hardware envelope

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TOP X-Y



FRONT X-Z



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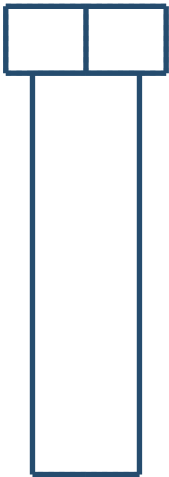
Nut, washer stack, grip, grade and preload unresolved; envelope only.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

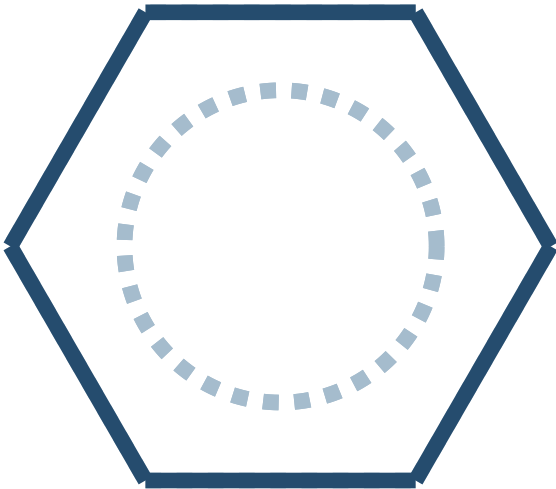
SIDE Y-Z



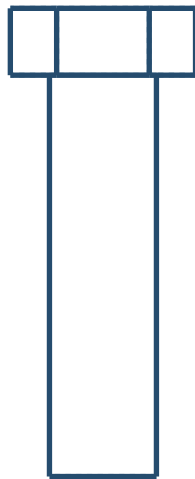
AB-G315-006 / M16 mounting hardware envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy after joint/preload calculation

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M16 is our candidate for Ø18 clearance holes, not a GALVI specified bolt.

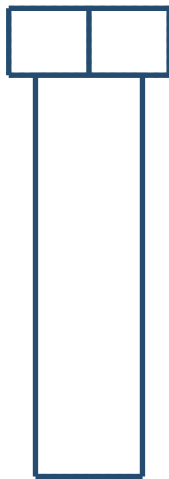
Nut, washer stack, grip, grade and preload unresolved; envelope only.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

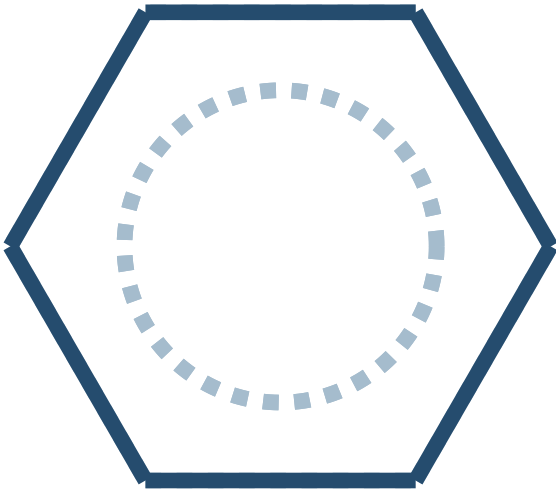
SIDE Y-Z



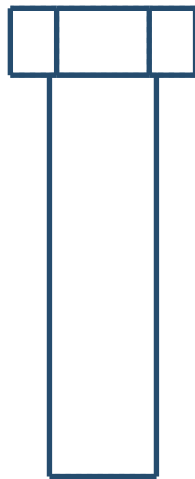
AB-G315-007 / M16 mounting hardware envelope

AI BRAKING / AB-G315 / C0 / 2026-09-07

TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy after joint/preload calculation

Overall X/Y/Z bounds: 27.70 / 23.99 / 70.00 mm

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SIDE Y-Z

